Class Marks Grid Analysis

Category B Practice - Part A, Problem 4

Problem summary

Calculate each student’s total, three subject averages, and the topper. Input marks are a 2D array with exactly three subject columns.

Java solution

```
public class Main {
 static void analyze(String[] names,int[][] marks){
  if(names.length==0||names.length!=marks.length)throw new IllegalArgumentException("Invalid rows");
  int rows=marks.length;double[] subjectTotals=new double[3];int best=0,bestTotal=Integer.MIN_VALUE;
  System.out.print("Totals ");
  for(int i=0;i<rows;i++){if(marks[i].length!=3)throw new IllegalArgumentException("Need 3 subjects");int total=
   for(int j=0;j<3;j++){total+=marks[i][j];subjectTotals[j]+=marks[i][j];}
   if(i>0)System.out.print(", ");System.out.print(names[i]+" "+total);if(total>bestTotal){bestTotal=total;best=i
  }
  System.out.println();System.out.printf("averages %.2f, %.2f, %.2f%n",subjectTotals[0]/rows,subjectTotals[1]/ro
  System.out.println("topper "+names[best]+" ("+bestTotal+")");
 }
 public static void main(String[] args){analyze(new String[]{"Asha","Ravi","Neha"},new int[][]{{78,85,90},{88,92
}
```

Sample output

```
Totals Asha 253, Ravi 259, Neha 230
averages 77.00, 82.33, 88.00
topper Ravi (259)
```

Complexity

For r students and c subjects, time is O(r*c). The subject totals require O(c) extra space; here c=3.